

A disproof of Conjecture 00000008524

The degree of the differential dimension polynomial

1 The conjecture and the reading

Conjecture: The differential dimension polynomial has rational coefficients; its degree is the number of variables and its leading coefficient is an integral multiple of $1/n!$.
(differential Hilbert polynomial)

Setting. Fix $n \geq 0$ differential indeterminates and $m \geq 0$ commuting derivations $\delta_1, \dots, \delta_m$ over $\mathbb{Q}$. The differential polynomial ring is $\mathbb{Q}\{y_1, \dots, y_n\} = \mathbb{Q}[y_{j,\theta}]$ with $j < n$, $\theta \in \mathbb{N}^m$ (Lean: `DRing n m`), $\delta_k y_{j,\theta} = y_{j,\theta+e_k}$ (`delta`). A *differential ideal* is an ideal stable under all δ_k (`IsDiffIdeal`); we only consider proper ones. Write $|\theta| = \sum_k \theta_k$ (`ord`) and $R_t = \mathbb{Q}[y_{j,\theta} : |\theta| \leq t]$. For a differential ideal I put

$$\omega_I(t) = \dim_{\text{Krull}} R_t / (I \cap R_t) \quad (\text{omega, Trunc}).$$

For a prime I with generic zero η this is the transcendence degree of $\mathbb{Q}(\eta_{j,\theta} : |\theta| \leq t)$ over $\mathbb{Q}$, i.e. Kolchin's dimension function. A *dimension polynomial* of I is $P \in \mathbb{Q}[X]$ with $\omega_I(t) = P(t)$ for all large t (`IsDimPoly`).

Readings of “the number of variables”. The number N is read as m (number of derivations) or as n (number of differential indeterminates); the same N is used in “leading coefficient an integral multiple of $1/N!$ ”. **Conjecture N** states: every proper differential ideal has a dimension polynomial P with $\deg P = N$ and $\text{lc}(P) = z/N!$ for some $z \in \mathbb{Z}$. (Rationality of the coefficients holds by $P \in \mathbb{Q}[X]$.) Convention: $\deg$ is Mathlib's `natDegree`, so constants have degree 0.

2 The refutation

Theorem 1. *For every $N : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}$ with $N(1,1) \neq 0$, Conjecture N is false (`C8524.not_conj`). In particular Conjecture fails for $N = m$ and for $N = n$ (`C8524.main : Claim`).*

Proof. Take $n = m = 1$ (ordinary case, one unknown y , $y_k = \delta^k y$). Let $\varphi : \mathbb{Q}\{y\} \rightarrow \mathbb{Q}[X]$ be the $\mathbb{Q}$ -algebra map with $y_0 \mapsto X$ and $y_k \mapsto 0$ for $k > 0$ (`phi`), and $W = \ker \varphi = [y']$ (`W`). It is prime (kernel to a domain, `W_prime`), proper, and differential: the composite $\varphi \circ \delta$ vanishes identically, by induction on polynomials ($\varphi(\delta y_k) = \varphi(y_{k+1}) = 0$ and the Leibniz rule), so $p \in W \Rightarrow \delta p \in W$ (`phi_delta, W_diff`).

Let $\psi_t : R_t \rightarrow \mathbb{Q}[X]$ be the restriction of φ . It is surjective (X is in the image), and $W \cap R_t = \ker \psi_t$ (`trunc_eq, psi_surj`). Hence $R_t / (W \cap R_t) \cong \mathbb{Q}[X]$ and $\omega_W(t) = \dim \mathbb{Q}[X] = \dim \mathbb{Q} + 1 = 1$ for every t (`omega_W`; Mathlib's `Polynomial.ringKrullDim_of_isNoetherianRing`).

If P were a dimension polynomial of W , then $P(t) = 1$ for all large t , so $P - 1$ has infinitely many roots and $P = 1$ (`Polynomial.eq_of_infinite_eval_eq`). Then $\deg P = 0 \neq N(1,1)$.

So the degree clause fails for W , and **Conjecture N** is false. For $N = m$ and $N = n$ we have $N(1,1) = 1$. □ □

3 Scope and Lean audit

Refuted: the clause “degree equals the number of variables” under the two readings $N = m$, $N = n$ (indeed any N with $N(1,1) \neq 0$), in the sense of Krull-dimension truncations above, with the witness a prime differential ideal. Not refuted: the clauses on rational coefficients and the $1/N!$ multiple (they hold for the witness), and a reading with $N(1,1) = 0$. Lean file **Conjecture8524/Basic.lean** (statement block followed by the proof); theorems **C8524.not_conj**, **C8524.main**. Axioms: only **propext**, **Classical.choice**, **Quot.sound**; no **sorry**.